

Tools — Shell

Tools for running shell commands and managing background tasks. `bash` commands pass through a 7-layer security model: safe-command whitelist, banned-command checks, denied patterns, directory-escape prevention, syntax validation, obfuscation detection, and user allowlist/denylist.

Tool	Description
<code>bash</code>	Execute a shell command (7-layer security).
<code>bash_reset</code>	Reset persistent shell state.
<code>bg</code>	Manage background shell tasks (spawn, list, wait, cancel).
<code>open</code>	Open a file/folder/URL via the desktop handler.

Use cases: running builds, tests, git operations, long-running tasks.

System instruction: “For simple commands use `bash` with the `command` parameter immediately. Do not describe what you will run.”

`bash`

Execute a shell command and return its stdout and stderr. Commands run in the agent’s working directory in a persistent shell.

Arguments

Argument	Type	Required	Description	Typical value
<code>command</code>	string	yes	Shell command to execute	"cargo test"
<code>timeout</code>	integer	no	Timeout in seconds	120 (default); up to 1500 for builds/tests

Example:

```
bash command="cargo test -- --nocapture" timeout=1500
bash command="cargo fmt"
```

`bash_reset`

Reset the persistent shell state for the current session, clearing saved working-directory changes and environment variables. No-op if no persistent state exists.

Arguments: none.

`bash_reset`

bg

Manage background shell tasks: spawn long-running commands that continue while the agent does other work, then inspect, follow, or stop them.

Arguments

Argument	Type	Required	Description	Typical value
action	enum	yes	spawn, list, status, output, tail, cancel, wait, cleanup	"spawn"
command	string	yes for spawn	Shell command to run in background	"cargo build"
task_id	string	yes for status, output, tail, cancel, wait	Background task id	"bg-3"
lines	integer	no	Lines returned by tail	20
status	enum	no	list filter: running / completed / failed / cancelled	—
limit	integer	no	Max tasks for list	50
timeout	integer	no	wait timeout in seconds	60
working_dir	string	no	Working directory for spawn	session cwd
completed_only	boolean	no	cleanup: only completed/failed/cancelled tasks	true
older_than_minimum_age	integer	no	cleanup: minimum task age	60

Example:

```
bg action="spawn" command="cargo build"
bg action="tail" task_id="bg-1" lines=20
bg action="wait" task_id="bg-1" timeout=300
```

open

Open or reveal a file, folder, or URL using the desktop default handler (xdg-open on Linux, open on macOS, start on Windows). May fail in headless or sandboxed contexts.

Arguments

Argument	Type	Required	Description	Typical value
target	string	yes	File path, folder path, or URL	"report.pdf"
action	enum	no	open (default), reveal (open parent directory), url (validate and open a URL)	"open"

Example:

```
open target="docs/howtos/tools/pdf/file-operations.md.pdf"
```